

IM3 Open Source Data Center Atlas

Summary

This dataset contains locations and attributes of existing data center facilities in the United States.

The dataset was developed through the Integrated Multisector, Multiscale Modeling (IM3) project, led by Pacific Northwest National Laboratory (PNNL) and supported by the U.S. Department of Energy Office of Science.

Data center locations were derived from OpenStreetMap (OSM) and augmented with additional geographic attributes including facility area, county, and state information. The dataset can be used to identify patterns of data center development and support infrastructure planning and geospatial analysis.

Database Information

Data were obtained from the IM3 Open Source Data Center Atlas repository maintained through the MSD-LIVE Data Repository.

Dataset record:

- IM3 Open Source Data Center Atlas
- DOI: 10.57931/2550666
- Published: 2025-06-06
- Version: v1

Data are available for download in:

- CSV format
- GeoPackage (GPKG) format

Dataset landing page:

<https://data.msdlive.org/records/p147s-4h760>

Interactive Atlas:

<https://immm-sfa.github.io/datacenter-atlas/>

Data were derived primarily from OpenStreetMap (OSM) data and enriched using U.S. Census Bureau state and county boundary datasets.

Data were accessed: 08 June 2025

CSV file was saved as:

IM3_OpenSource_DataCenter_Atlas_Raw.csv

Data Content Information

Description

This dataset contains locations of existing data center facilities in the United States. Data center locations were derived from OpenStreetMap (OSM), a crowd-sourced database. Additional variables were generated through spatial processing, including facility area (square feet), associated county, and state.

Coordinate Reference System

Geospatial data are provided in:

- WGS84 (EPSG:4326) coordinate reference system.

Available Layers

The GeoPackage version contains three layers:

Layer	Description
point	Data centers represented as point locations
building	Individual data center building footprints
campus	Data center campus footprints

A facility may appear in more than one layer. Some facilities are represented only as points, while others include building and/or campus polygons.

Variables

Variable	Description
id	Unique facility identifier
state	State name
state_abb	Two-letter state abbreviation
state_id	State FIPS identifier
county	County name
county_id	County FIPS identifier
ref	OpenStreetMap reference code
operator	Facility operator or owner
name	Facility name
sqft	Facility area in square feet (building and campus layers only)
lat	Latitude of feature centroid
lon	Longitude of feature centroid
type	Spatial representation ("point", "building", or "campus")
geometry	Spatial geometry (GPKG only; not included in CSV download)

Variables marked as OSM-derived originate directly from OpenStreetMap records.

Usage Notes

Validation of OSM-derived data center locations is ongoing.

Important considerations:

- Some facilities are represented as both campuses and individual buildings.
- Overlapping geometries are retained in the dataset.
- Facilities crossing county boundaries may appear as multiple records.
- County-specific records retain identical facility identifiers but different county attributes.
- Coverage is dependent on OpenStreetMap contributions and may be incomplete in some areas.
- The dataset is expected to be updated periodically as new facilities are identified and OSM data improve.

Attribution

Data center locations were derived from OpenStreetMap and are distributed under the Open Database License (ODbL).

State and county boundary information was obtained from the U.S. Census Bureau 2024 geographic boundary files.

Citation

Mongird, K., Thurber, T., Vernon, C., Burleyson, C., Akdemir, K. Z., & Rice, J. (2025).

IM3 Open Source Data Center Atlas (Version 1).

MSD-LIVE Data Repository.

DOI: <https://doi.org/10.57931/2550666>

License

The IM3 Open Source Data Center Atlas is distributed under the Open Database License (ODbL) v1.0.

License information:

<http://opendatacommons.org/licenses/odbl/1.0/>

Naming Conventions and File Formats

Files are named according to the following convention:

```
database_name_datatype_details_stage.format
```

where:

database_name refers to the source database

datatype describes the data content

details provides additional descriptive information

stage refers to the data processing stage (e.g., raw, cleaned, processed)

format refers to the file format (e.g., .csv, .gpkg, .txt)

Additional Information and Support

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Pacific Northwest National Laboratory (PNNL)

Project:

Integrated Multisector, Multiscale Modeling (IM3) Project

For additional information, consult the dataset landing page:

<https://data.msdlive.org/records/65g71-a4731>

or the interactive atlas:

<https://immm-sfa.github.io/datacenter-atlas/>